In the First World War it made sense to send millions of raw conscripts to charge machine guns and die in their thousands. Their individual skills mattered little. Today, despite the shortage of drone operators and data analysts, the US Air Force is unwilling to fill the gaps with Walmart dropouts. You wouldn't like an inexperienced recruit to mistake an Afghan wedding party for a high-level Taliban conference.

Consequently, despite the appearance of many new human jobs, we might nevertheless witness the rise of a new 'useless' class. We might actually get the worst of both worlds, suffering simultaneously from high unemployment and a shortage of skilled labour. Many people might share the fate not of nineteenth-century wagon drivers – who switched to driving taxis – but of nineteenth-century horses, who were increasingly pushed out of the job market altogether.[15]

In addition, no remaining human job will ever be safe from the threat of future automation, because machine learning and robotics will continue to improve. A forty-year-old unemployed Walmart cashier who by dint of superhuman efforts manages to reinvent herself as a drone pilot might have to reinvent herself again ten years later, because by then the flying of drones may also have been automated. This volatility will also make it more difficult to organise unions or secure labour rights. Already today, many new jobs in advanced economies involve unprotected temporary work, freelancing and one-time gigs.[16] How do you unionise a profession that mushrooms and disappears within a decade?

Similarly, human–computer centaur teams are likely to be characterised by a constant tug of war between the humans and the computers, instead of settling down to a lifelong partnership. Teams made exclusively of humans – such as Sherlock Holmes and Dr Watson – usually develop permanent hierarchies and routines that last decades. But a human detective who teams up with IBM's Watson computer system (which became famous after winning the US TV quiz show *Jeopardy!* in 2011) will find that every routine is an invitation for disruption, and every hierarchy an invitation for revolution. Yesterday's sidekick might morph into tomorrow's superintendent, and all protocols and manuals will have to be rewritten every year.[17]

A closer look at the world of chess might indicate where things are heading in the long run. It is true that for several years after Deep Blue defeated Kasparov, human–computer cooperation flourished in chess. Yet in recent years computers have become so good at playing chess that their human collaborators lost their value, and might soon become utterly irrelevant.

On 7 December 2017 a critical milestone was reached, not when a computer defeated a human at chess – that's old news – but when Google's AlphaZero program defeated the Stockfish 8 program. Stockfish 8 was the world's computer chess champion for 2016. It had access to centuries of